Malhar Gudekar

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Education

University of Illinois Urbana Champaign August 2024 - May 2026
Master of Science, Information Management GPA: 3.8/4
• Coursework: Data Warehousing & Business Intelligence, Information Management, Information Consulting, Methods of Data Science

University of Mumbai June 2019 - May 2023
Bachelor of Engineering, Computer Science GPA: 9.25/10
• Coursework: DBMS, Big Data Analysis, Data Warehousing and Mining, Project Management, NLP, Machine Learning, Social Media Analytics

Skills

• **Machine Learning / Statistics:** Classification, Regression, Feature Engineering, Model Evaluation, Precision, Recall, Accuracy, Statistical Analysis, Predictive Analytics, Data Validation
• **Programming / Querying:** Python, SQL, Pandas, NumPy, PySpark, PostgreSQL, Neo4j, Scikit-learn, LightGBM
• **Visualization / Reporting:** Power BI, Tableau, DAX, Excel, VBA, Dashboard Development, Data Visualization
• **Data / Cloud / Tools:** AWS S3, EC2, Redshift, EKS, FastAPI, Docker, Git, Kafka, pgVector, GCP

Work Experience

University of Illinois Urbana Champaign May 2025 - Present
Research Assistant - iSchool
- Designed and implemented end-to-end **machine learning** pipeline to ingest, process, and analyze wearable data from 100+ users, combining scalable **data ingestion**, **feature engineering**, and model-ready transformations to enable longitudinal behavioral insights and downstream analytics
- Developed and optimized **PostgreSQL** data models, indexing strategies, and query execution plans for high-volume datasets, improving query performance by **38%** while enabling efficient feature extraction for machine learning workflows
- Developed, validated, and deployed ML-based retrieval pipeline with performance benchmarking and monitoring, improving multi-turn response accuracy by 43% across longitudinal behavioral datasets

Research Assistant - CHI AI in Infodemic Management January 2025 - May 2025
- Developed NLP and statistical modeling pipelines to classify health misinformation from large-scale social media data, applying feature extraction, model validation, and iterative optimization to reduce latency by 41%
- Built distributed **data pipelines** using **Kafka** and **Neo4j** Streams to enable **scalable ingestion**, transformation, and real-time analysis of high-volume unstructured datasets
- Designed production-grade inference **APIs** and deployment workflows to support reproducibility, monitoring, and consistent evaluation of machine learning outputs

Business Intelligence Group August 2025 - December 2025
Technical Consultant
- Built and deployed **Retrieval-Augmented Generation** (RAG) system using embeddings and **vector search**, combining data pipelines, feature engineering, and retrieval optimization to improve document retrieval efficiency by 64%
- Designed scalable **ETL** and feature pipelines using **FastAPI** and **PostgreSQL** to transform unstructured data into structured formats suitable for analytics and machine learning use cases
- Optimized indexing strategies and data processing workflows to enable low-latency querying and scalable deployment of AI-driven systems

Swift Mobil Software Solutions Provider July 2023 - October 2023
Data Analyst
- Analyzed large-scale operational datasets using **Python** and **PySpark** to identify trends, anomalies, and performance issues, combining exploratory analysis and root cause investigation to support data-driven decision-making
- Built **Power BI** dashboards to track key performance indicators and improve visibility into operational performance, reducing reporting time by 40%

Pscope Technologies Pvt. Ltd. January 2023 - June 2023
Data Analyst
- Developed and maintained Power BI dashboards using **SQL** and **DAX**, transforming raw data into actionable insights for business decision-making across multiple domains.
- Performed data validation, cleaning, and transformation on large multi-source datasets using **SQL** and **Excel**, improving data accuracy by over 30% and reducing reporting inconsistencies, enabling faster and more reliable decision-making across sales, finance, and operations teams

Stu/Dio at Illinois August 2025 - Present
Project Manager
- Led a **cross-functional** team of six to deliver and scale Deepcover game enhancements by leveraging user data, performance metrics, and feedback analysis to drive feature prioritization and improve games outcomes

Projects

Scalable ML System for Customer Risk Prediction
- Built **LightGBM**-based machine learning model to **predict customer credit risk** from **time-series transaction data**, combining feature engineering, classification modeling, and evaluation techniques to identify high-risk and low-risk segments.
- Designed and ran controlled model experiments, engineering behavioral features and iterating on classification thresholds through systematic validation — improving credit risk prediction precision by 20%
- Evaluated **model performance** using precision, recall, and accuracy, and designed a decision framework to translate predictions into actionable credit limit recommendations.